

Accelera Project Workflow

1. Auto Preprocessing

Data profiling
and type detection

Task-specific
preprocessing

Tabular / Text / Image
preparation

Processed data
+ reports
+ preprocessors

2. Accelera AutoML

Statistics
and metrics

Candidate models
+ preprocessing choices

Search /
optimization

Best model
+ score
+ run history

Predictions
+ executed graph
+ reports

6. Deployment

Build service
and API

Publish endpoint
+ optional UI

Deployed model URL
/ service

4. Parallelizer

Normalize input
Python to C++ if needed

Loop extraction
Clang AST

Classifier
+ safety checks

5. Benchmark Platform

Prediction CSV
+ optional repo URL

Compare with
true labels

Score
+ ranking

3. Accelera Pipe

optimize custom method

Graph of nodes
preprocessors, branches,
merge, I/O

Branch / merge
execution

return optimized callable

Parallelized C++
or optimized callable

Dataset /
processed data

Dataset

Selection switch
selected model or pipeline

Saved pipeline
or model

User

Benchmark data
train / test / truth

User workflow

C/C++ or Python
source / function

Parallel independent
paths

E2E output